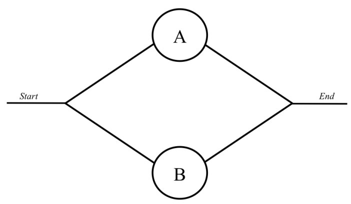
Stat 641: Section 1
White
Fall 2021
Exam 1

1. (5 points) Let $\mathcal{F}$ a σ-algebra on some sample space S. Prove that if any $A_1, A_2, ... \in \mathcal{F}$, then $\bigcap\limits_{n=1}^{\infty} A_n \in \mathcal{F}$ (closed under countable intersection).

   Let $A_1, A_2, ..., \in \mathcal{F}$. Because $\mathcal{F}$ is closed under complement, $A_1^c, A_2^c, ..., \in \mathcal{F}$. Because $\mathcal{F}$ is closed under countable union, $\bigcup\limits_{n=1}^{\infty} A_n^c \in \mathcal{F}$. Then, again because $\mathcal{F}$ is closed under complement, $\left[\bigcup\limits_{n=1}^{\infty} A_n^c\right]^c = \bigcap\limits_{n=1}^{\infty} A_n \in \mathcal{F}$, by DeMorgan's law.

2. Consider the system of two components, A and B shown below, where component B is higher quality than component A. This is a parallel system, meaning the system will function if A functions or B functions. Suppose A functions with probability p. If A functions, then B functions with probability $r > p$. If A has failed, then B functions with probability p.



   (a) (3 points) What is the probability that B functions?

   $P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) = rp + p(1-p) = rp + p - p^2 = p(1 + r - p)$

   (b) (3 points) What is the probability that the system functions?

   $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(B|A)P(A) = p + rp + p - p^2 - rp = 2p - p^2 = p(2-p)$

   (c) (4 points) If B does not function, what is the probability that A does?

   $P(A|B^c) = \frac{P(B^c|A)P(A)}{P(B^c|A)P(A)+P(B^c|A^c)P(A^c)} = \frac{(1-r)p}{(1-r)p+(1-p)(1-p)} = \frac{(1-r)p}{(1-r)p+(1-p)^2} = \frac{p-rp}{1+p^2-pr-p}$

3. Let $X \sim \text{Beta}(a, b)$. The pdf of X is

$$f(x) = \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1} & 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

   where $a > 0$ and $b > 0$.

   (a) (5 points) Find $\mathbb{E}\left(\frac{1-X}{X}\right)$ without finding the distribution of $\frac{1-X}{X}$.

$$\begin{aligned}
\mathbb{E}\left(\frac{1-X}{X}\right) &= \int_0^1 \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{1-x}{x} x^{a-1}(1-x)^{b-1} dx \\
&= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 x^{a-1-1}(1-x)^{b+1-1} dx \\
&= \frac{\Gamma(a+b)\Gamma(a-1)\Gamma(b+1)}{\Gamma(a)\Gamma(b)\Gamma(a+b)} \int_0^1 \frac{\Gamma(a+b)}{\Gamma(a-1)\Gamma(b+1)} x^{a-1-1}(1-x)^{b+1-1} dx
\end{aligned}$$

$$= \frac{\Gamma(a+b)\Gamma(a-1)\Gamma(b+1)}{\Gamma(a)\Gamma(b)\Gamma(a+b)}, \qquad \text{if } a > 1$$
$$= \frac{b}{a-1}, \qquad \text{if } a > 1, \text{ by properties of } \Gamma(\cdot).$$

(b) (5 points) Derive the PDF $f_Y(y)$ for $Y = -\log\left(\frac{X}{1-X}\right)$, where log is the natural logarithm.

- Support $Y \in \mathcal{Y} = (-\infty, \infty)$
- $X = g^{-1}(Y) = \frac{e^{-Y}}{1+e^{-Y}} = \frac{1}{1+e^{Y}}$
- $\left|\frac{dg^{-1}(Y)}{dY}\right| = \frac{e^{-Y}}{(1+e^{-Y})^2} = \frac{e^{Y}}{(1+e^{Y})^2}$
-

$$f_Y(y) = \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{e^{-y}}{(1+e^{-y})^2} \left(\frac{e^{-y}}{1+e^{-y}}\right)^{a-1} \left(\frac{1}{1+e^{-y}}\right)^{b-1} & y \in \mathbb{R} \\ 0 & \text{otherwise} \end{cases}$$
$$= \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \left(\frac{e^{-y}}{1+e^{-y}}\right)^{a} \left(\frac{1}{1+e^{-y}}\right)^{b} & y \in \mathbb{R} \\ 0 & \text{otherwise} \end{cases}$$

4. (6 points) Consider a random variable with pdf is

$$f(x) = \begin{cases} 2x & x \in (0,1), \\ 0 & \text{otherwise.} \end{cases}$$

Derive the CDF of $F_Y(y)$ for $Y = X(1-X)$ and prove that it is a valid CDF.

First, note that $F_X(x) = x^2$ for $x \in (0,1)$, 0 when $x < 0$, and 1 when $x > 1$.

$$\begin{aligned}
F_Y(y) &= P(Y \le y) = P(X(1-X) \le y) = P(X - X^2 \le y) \\
&= P(-(X^2 - X + 1/4) + 1/4 \le y) = P(-(X-1/2)^2 + 1/4 \le y) \\
&= P(-(X-1/2)^2 \le y - 1/4) = P((X-1/2)^2 > -(y-1/4)) \\
&= 1 - P((X-1/2)^2 \le (1/4 - y)) \\
&= 1 - P(1/2 - \sqrt{1/4-y} \le X \le 1/2 + \sqrt{1/4-y}) \\
&= 1 - F_X(1/2 + \sqrt{1/4-y}) + F_X(1/2 - \sqrt{1/4-y}) \\
&= 1 - (1/2 + \sqrt{1/4-y})^2 + (1/2 - \sqrt{1/4-y})^2 \\
&= 1 - 1/4 - \sqrt{1/4-y} - (1/4-y) + 1/4 - \sqrt{1/4-y} + (1/4-y) \\
&= 1 - 2\sqrt{1/4-y} = \begin{cases} 0 & y < 0 \\ 1 - \sqrt{1-4y} & y \in [0, 1/4) \\ 1 & y \ge 1/4 \end{cases}
\end{aligned}$$

It is a valid CDF because

- The limits on this one are really easy because it is already 1 for all $y > 1/4$ and 0 for all $y < 0$. So, $\lim_{y\to-\infty} F_Y(y) = 0$ and $\lim_{y\to\infty} F_Y(y) = 1$.
- Similarly, the CDF is not only right-continuous, it is completely continuous. For all $y \in [0, 1/4]$, $\lim_{a\to y} F_Y(a) = F_Y(y)$ (that closed right-bracket is intentional). In other words, even at points where we may suspect discontinuity ($y \in \{0, 1/4\}$), the function $1-\sqrt{1-4y}$ meets 0 and 1 at 0 and $1/4$, respectively.

- This CDF is non-decreasing outside of $(0, 1/4]$ because it is completely flat (0 or 1). We'll take a derivative to show that it is increasing over $(0, 1/4]$. $\frac{d}{dy}F(y) = \frac{d}{dy}1 - \sqrt{1-4y} = \frac{2}{\sqrt{1-4y}} \geq 0$ for $y \in (0, 1/4]$. Thus, it is nondecreasing everywhere.

5. (5 points) Consider a random variable with pdf is

$$f(x) = \begin{cases} \frac{n(n-1)}{2^n} x^{n-2}(2-x) & x \in (0,2), \\ 0 & \text{otherwise,} \end{cases}$$

where $n \in \{2, 3, ...\}$ is an unknown parameter. Using the fact that this in an exponential family, find the expected value of $\ln(X)$.

There was a small error in the problem. We need n to be at least 2. Note that we can write this pdf as

$$f(x) = \frac{n(n-1)}{2^n}(2-x)x^{-2}e^{n\log(x)}\mathbf{1}_{(0,2)}(x).$$

So, $\eta = n$ or $n-2$, depending on how you wrote this. Now, because $n = \eta$,

$$\begin{aligned} \mathbb{E}(\log(x)) &= -\frac{\partial}{\partial n}\log(c^\star(n)) \\ &= -\frac{\partial}{\partial n}\left[-n\log(2) + \log(n) + \log(n-1)\right] \\ &= \log(2) - \frac{1}{n} - \frac{1}{n-1} \end{aligned}$$

6. With $a > 0$ and $b > 0$, the pdf of a Gamma(a, b) random variable is

$$f(x) = \begin{cases} \frac{1}{\Gamma(a)b^a} x^{a-1}e^{-x/b} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) (5 points) Let $X \sim \text{Gamma}(\alpha n, \beta/n)$. Find the MGF of X.

$$\begin{aligned} \mathbb{E}(e^{tX}) &= \int_0^\infty e^{tx}\frac{1}{\Gamma(\alpha n)(\beta/n)^{n\alpha}}x^{\alpha n-1}e^{-x/(\beta/n)}dx \\ &= \int_0^\infty \frac{1}{\Gamma(\alpha n)(\beta/n)^{n\alpha}}x^{\alpha n-1}e^{-x(n/\beta - t)}dx \\ &= \frac{\left[(n/\beta-t)^{-1}\right]^{\alpha n}}{(\beta/n)^{n\alpha}}\int_0^\infty \frac{1}{\Gamma(\alpha n)\left[(n/\beta-t)^{-1}\right]^{\alpha n}}x^{\alpha n-1}e^{-x/(n/\beta-t)^{-1}}dx \\ &= \frac{\left[((n-t\beta)/\beta)^{-1}\right]^{\alpha n}}{(\beta/n)^{n\alpha}} \qquad \text{if } t < \frac{n}{\beta} \\ &= \frac{[\beta/(n-t\beta)]^{\alpha n}}{(\beta/n)^{n\alpha}} = \left(\frac{n}{n-t\beta}\right)^{\alpha n} = \left(\frac{1}{1-t\beta/n}\right)^{\alpha n}, \text{ if } t < n/\beta \end{aligned}$$

(b) (3 points) What is $\lim_{n\to\infty} M_X(t)$? (You may assume $\lim_{n\to\infty}(c_n)^d = c^d$ given that $\lim_{n\to\infty} c_n = c$) Bonus: What does this mean about the limiting distribution of X?

With a little bit of rewriting,

$$\mathbb{E}(e^{tX}) = \left(\frac{1}{1-t\beta/n}\right)^{\alpha n} = \left(\frac{n-t\beta}{n}\right)^{-\alpha n}$$

$$= \left[\left(1 + \frac{-t\beta}{n} \right)^n \right]^{-\alpha}$$

So, $\lim_{n\to\infty} \left[(1 + \frac{-t\beta}{n})^n \right]^{-\alpha} = \left[e^{-t\beta} \right]^{-\alpha} = e^{t\alpha\beta}$. Now, there are no restrictions on t.

Bonus: This means that the limiting distribution is a point mass at $\alpha\beta$ (the mean of the Gamma distribution).

7. Consider a random variable X with the following triangle distribution over $\mathcal{X} = (-3, 4)$,

$$f(x) = \begin{cases} \frac{x+3}{7} & -3 < x \leq -1, \\ \frac{2(4-x)}{35} & -1 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) (4 points) What is the probability that $X > 1$ given that X is positive.

$$P(X > 1 | X > 0) = \frac{P(X > 1 \cap X > 0)}{P(X > 0)} = \frac{P(X > 1)}{P(X > 0)}$$
$$= \frac{\int_1^4 \frac{2(4-x)}{35} dx}{\int_0^4 \frac{2(4-x)}{35} dx} = \frac{9/35}{16/35} = \frac{9}{16}$$

(b) (8 points) Find $f_Y(y)$ for $Y = X^2$. Okay, this problem is a bit of a pain. $X = g^{-1}(Y) = -\sqrt{Y}$ when $X < 0$, and $X = g^{-1}(Y) = \sqrt{Y}$ when $X \geq 0$. In either case, $\left| \frac{dX}{dy} \right| = \frac{1}{2\sqrt{y}}$. Then, there are two more challenges: (1) we have to find regions in $\mathcal{X}$ that provide the same support for Y while (2) we account for the definition of the pdf changing at -1. So, we need the following:

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} \left[\frac{2(4-\sqrt{y})}{35} + \frac{2(4+\sqrt{y})}{35} \right] = \frac{8}{35\sqrt{y}} & y \in [0, 1] \\ \frac{1}{2\sqrt{y}} \left[\frac{3-\sqrt{y}}{7} + \frac{2(4-\sqrt{y})}{35} \right] = \frac{23-7\sqrt{y}}{70\sqrt{y}} & y \in (1, 9] \\ \frac{1}{2\sqrt{y}} \left[\frac{2(4-\sqrt{y})}{35} \right] = \frac{4-\sqrt{y}}{35\sqrt{y}} & y \in (9, 16) \\ 0 & \text{otherwise} \end{cases}$$

8. Assume that the average length of a female great white shark is 16 ft.

(a) (2 points) Without making more assumptions, provide an upper bound on the probability that a random female great white shark is over 20 feet long.
$P(X > 20) \leq \frac{\mathbb{E}(X)}{20} = \frac{16}{20} = 4/5$

(b) (2 points) Now suppose that the standard deviation of the length is 4/3. Now, give a lower bound on the probability that a random female great white shark is between 12 and 20 feet.
$P(12 < X < 20) = P(|X - 16| < 4) = P(|X - 16| < 3\sigma) \geq 1 - \frac{1}{3^2} = 8/9$